

Chapter 4 - Operators Exercises

1. Write Java expressions using the following arithmetic operators: +, -, *, /, %. Provide examples and expected outputs.
2. Explain the behavior of the modulus operator (%) with negative and floating-point numbers. Provide examples.
3. Java does not have an exponent operator. How can you compute x raised to the power y in Java?
4. Explain type casting in Java. Provide examples of both implicit and explicit casting.
5. Write a Java program that demonstrates the use of compound assignment operators (+=, -=, *=, /=, %=).
6. Write a Java program to demonstrate the difference between prefix and postfix increment/decrement operators. Use print statements to show the results.
7. Given x = 5 and y = 8, evaluate the following expressions:
 - a) x == y
 - b) x != y
 - c) x > y
 - d) x <= y
8. Write a Java expression that returns true if a year is a leap year. A year is a leap year if it is divisible by 4 but not by 100, or divisible by 400.
9. Write a Java program that lets users input two numbers and performs addition, subtraction, multiplication, and division.
10. Write a Java program that lets users input scores of three subjects, calculates the GPA, and prints the result.